

A Necessary and Sufficient Pumping Lemma for Regular Languages

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1. Introduction

A well known necessary and sufficient condition to guarantee that a language is regular is provided by Nerode's Theorem [3]. However, the "pumping lemma" is often a more useful tool in demonstrating properties of regular languages. The question has been raised [1] whether there is a necessary and sufficient pumping lemma. In this note we answer the question in the affirmative.

2. Pumping Lemma

Theorem. Let Σ be a finite alphabet. A language $L \subseteq \Sigma^*$ is regular iff

(*) $\exists k \in \mathbb{N}$ such that $\forall y \in \Sigma^*$ with $|y| = k$ $\exists u, v, w \in \Sigma^*$ satisfying $y = uvw$, $|v| > 0$ and $\forall z \in \Sigma^*, \forall i \in \mathbb{N}$
 $yz \in L \Rightarrow uv^i wz \in L$ and $yz \notin L \Rightarrow uv^i wz \notin L$.

Proof. ($\Rightarrow$) Let $M = (Q, \Sigma, \delta, q_0, F)$ be a deterministic finite state recognizer (fsr) for L where Q is the state set, $\delta: Q \times \Sigma \rightarrow Q$ is the state transition function, $q_0 \in Q$ is the initial state and $F \subseteq Q$ is the set of final states. We let δ' denote the extension of δ to domain $Q \times \Sigma^*$ in the usual way. For $y \in \Sigma^{|Q|}$ the path taken by y from q_0 to $\delta'(q_0, y)$ must contain a loop. It is easy to see that for $k = |Q|$, (*) holds if we use any decomposition of y as $y = uvw$, where v is a subword of y that labels the arcs of one of the loops in the path. $\square$

($\Leftarrow$) While the result follows fairly easily from Nerode's Theorem, we will give a direct proof for those not familiar with Nerode's Theorem. If (*) is true then construct a fsr $M = (Q, \Sigma, \delta, q_0, F)$ that recognizes L as follows. (Note that δ' is defined as above.) The size of Q is $1 + |\Sigma| + |\Sigma|^2 + \dots + |\Sigma|^{k-1}$ and for distinct words $\ell, \ell' \in \Sigma^*$ of size less than k , $\delta'(q_0, \ell) \neq \delta'(q_0, \ell')$. Thus the fsr viewed as a graph is a complete k -ary tree, if we ignore the arrows that emanate from the leaves of this tree (which we have yet to describe). A state q is a final state iff the single word ℓ of length less than k such that $\delta'(q_0, \ell) = q$ has the property $\ell \in L$. To complete the specification of M , all that must be presented is $\delta(q, \sigma)$ where q is one of the states at a leaf of the tree and $\sigma \in \Sigma$.

Fix a state q which is at a leaf of the tree. The state has an associated word ℓ of length exactly $k-1$. To determine $\delta(q, \sigma)$ examine the word $y = \ell\sigma$. Since $|y| = k$, by (*) there must be words uvw such that $y = uvw$ and for any $z \in \Sigma^*$ and $i \in \mathbb{N}$, $yz \in L \Rightarrow uv^i wz \in L$ and $yz \notin L \Rightarrow uv^i wz \notin L$. Define $\delta(q, \sigma) = \delta'(q_0, uw)$. In this manner we may define $\delta(q, \sigma)$ for each state, q , at the leaf of the tree.

Now that we have given the complete specification of M , it remains only to verify that M accepts L . That is, we must show that for $\ell \in \Sigma^*$, $\ell \in L$ iff $\delta'(q_0, \ell) \in F$.

Lemma 1. Let $|y| = k$, and $y = uvw$ as in the statement of the Theorem. Then $\forall z \delta'(q_0, yz) = \delta'(q_0, uwz)$.

Proof. By construction $\delta'(q_0, y) = \delta'(q_0, uw)$. $\square$

Lemma 2. For $\ell \in \Sigma^*$, $\ell \in L$ iff $\delta'(q_0, \ell) \in F$.

Proof. Induction on $|\ell|$.

Basis. If $|\ell| < k$ then Lemma 2 follows from the way that final states were defined for M .

Inductive Step. Assume that we have proved Lemma 2 for all words of length less than n and let $|\ell| = n$. Since $n \geq k$ we can write $\ell = yz$ where $|y| = k$ and $y = uvw$ as in the Theorem. Now, $\ell \in L \Rightarrow uvwz \in L$ by the hypothesis of the Theorem where $|uvwz| < |\ell|$. But $uvwz \in L$ implies (by induction) that $\delta'(q_0, uvwz) \in F$. By Lemma 1, $\delta'(q_0, uvwz) = \delta'(q_0, \ell)$, and thus $\delta'(q_0, \ell) \in F$. Similarly, we may show that $\ell \notin L \Rightarrow \delta'(q_0, \ell) \notin F$. $\square$

Thus we have found a finite state recognizer for any language L satisfying (*). This proves the Theorem. $\square$

3. Other Necessary and Sufficient Conditions

While the Theorem of Section 2 relates regularity with a specific "pumping" condition, in fact it is easy to see that there are many other necessary and sufficient conditions. For example, there are seemingly stronger hypotheses than (*) which are true for regular languages, and thus also provide necessary and sufficient conditions. An example is the assertion:

$\exists k \in \mathbb{N}$ such that $\forall x, y \in \Sigma^*$ with $|y| \geq k$ $\exists u, v, w \in \Sigma^*$ satisfying $y = uvw$, $|v| > 0$, and $\forall z \in \Sigma^*, \forall i \in \mathbb{N}$

$$xyz \in L \Rightarrow xuv^i wz \in L \text{ and } xyz \notin L \Rightarrow xuv^i wz \notin L$$

We remark that a particular pumping lemma conjectured in [1] to be necessary and sufficient fails to be sufficient. The condition is:

(**) $\exists k \in \mathbb{N}$ such that $\forall x, y, z \in \Sigma^*$ with $|y| = k$ $\exists u, v, w \in \Sigma^*$ satisfying $y = uvw$, $|v| > 0$ and $\forall i \in \mathbb{N}$

$$xyz \in L \Rightarrow xuv^i wz \in L \text{ and } xyz \notin L \Rightarrow xuv^i wz \notin L.$$

It has recently been shown by J. Beauquier [2] that (**) is not sufficient for a specific context-free language over an eight letter alphabet.

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References

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3. Nerode A. "Linear automaton transformations," *Proc. Amer. Math. Soc.*, 9, 541-544.